# Zhenyu Fan

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## Education

- **Peking University** – Beijing, China — Sep 2023 - June 2027 (expected)
  PhD in Pure Mathematics
  *Advisors: Prof. Yu Yuan and Prof. Yuguang Shi; formerly advised by Prof. Qing Han*
- **Xi'an Jiaotong University** – Xi'an, China — Sep 2019 - June 2023
  Bachelor, Honors Science Program in Mathematics
  *Advisor: Prof. Dongsheng Li*
- **Xi'an Jiaotong University** – Xi'an, China — Sep 2017 - June 2019
  Young Gifted Class

## Research Interest

- **Nonlinear Elliptic Equations & Geometric Analysis**
  Especially minimal surfaces, Monge-Ampère equations, special Lagrangian equations and other fully nonlinear equations

## Research Papers

[8] **Zhenyu Fan**. Perturbation of special Lagrangian cones near infinity. *In preparation.* (2026)

[7] **Zhenyu Fan**, Ravi Shankar. Alexandrov regularity for solutions to fully nonlinear curvature equations . *In preparation.* (2026)

[6] **Zhenyu Fan**. Geometric regularity for semi-convex viscosity solutions to the special Lagrangian equation. *Submitted, arXiv:2609.07458.* (2026)

[5] **Zhenyu Fan**, Caiyan Li, Zhehui Wang. Failure of interior Hölder and gradient estimates for the subcritical special Lagrangian equation. *Submitted, arXiv:2609.01531.* (2026)

[4] **Zhenyu Fan**, Ravi Shankar. Interior estimates and regularity for the scalar curvature equation in dimension 4. *Submitted, arXiv:2608.22748.* (2026)

[3] **Zhenyu Fan**. Hessian estimates for the sigma-2 equation with variable right-hand side terms in dimension 4. *Advances in Mathematics* Volume 494 (2026)

[2] **Zhenyu Fan**. A generalization of Savin's small perturbation theorem for fully nonlinear elliptic equations and applications. *Calculus of Variations and Partial Differential Equations* **65**, 182 (2026)

[1] **Zhenyu Fan**, Yugao Ouyang. Asymptotic behavior of exterior minimal graphs in half spaces. *The Journal of Geometric Analysis* **35**, 247 (2025)

## Awards & Honors

- **Presidential Scholarship**, Peking University — 2026
  **China National Scholarship**, Ministry of Education of the People's Republic of China
- **Zhou Yulin Scholarship**, Peking University — 2025
- **Award for Scientific Research**, Peking University — 2024
  **Pingfanghe Investment Scholarship**, Peking University
- **Outstanding Graduates**, Xi'an Jiaotong University — 2023

## INVITED TALKS

3. **Title**: On the sigma-2 equation and the scalar curvature equation in dimension 4. April 9, 2026
   Seminar, Academy of Mathematics and Systems Science, CAS, Beijing, China.

2. **Title**: On the sigma-2 equation and the scalar curvature equation in dimension 4. March 6, 2026
   Seminar, Xi'an Jiaotong University, Xi'an, China.

1. **Title**: On the sigma-2 equation and the scalar curvature equation in dimension 4. December 3, 2025
   Doctoral Forum on Elliptic PDEs, Fudan University, Shanghai, China.

## TEACHING EXPERIENCE

**Peking University**, Teaching assistant
Independently prepared and led recitation sections for undergraduate calculus courses.

- Advanced Math A Fall 2026
- Advanced Math A Spring 2026
- Advanced Math A Fall 2025
- Advanced Math B Spring 2025
- Advanced Math B Fall 2024
- Advanced Math B Spring 2024
- Advanced Math C Fall 2023

## ACADEMIC SERVICES

**Referee for several journals** including: *Calculus of Variations and Partial Differential Equations*